

ZINONG YANG

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EDUCATION AND APPOINTMENT

Stanford University, Stanford, CA Post-doctoral Associate in Bioengineering	<i>Sep 2026 - Current</i>
Massachusetts Institute of Technology, Cambridge, MA Post-doctoral Fellow in RLE and IMES	<i>June 2025 - Aug 2026</i>
Massachusetts Institute of Technology, Cambridge, MA Graduate Visiting Student in Research Laboratory of Electronics and the Institute of Medical Engineering Systems	<i>June 2023 - June 2025</i>
Boston University, Boston, MA Ph.D. in Computational Neuroscience	<i>Sep 2019 - June 2025</i>
University of California, San Diego, La Jolla, CA Bachelor of Science in Cognitive Neuroscience; Minor: Philosophy Honors degree: Cum Laude	<i>Sep 2014 - June 2017</i>

PUBLICATIONS

- [1] **Yang, Z.**, S. Williams, and L. Lewis, "Attentional lapses emerge from interactions between global arousal state and local cortical dynamics," *In prep*, 2026.
- [2] E. Beldzik, N. Cicero, D. Gomez, **Yang, Z.**, J. Iglesias, B. Edlow, and L. Lewis, "A hypothalamic-brainstem activity sequence underlies arousal fluctuations during daytime drowsiness," *under revision in Nature Neuroscience*, 2026.
- [3] **Yang, Z.** and L. Lewis, "B³-net: Brain-body-behavior model of attention fluctuations in humans," *Submitted*, 2026.
- [4] J. Levitt, X. Zeng, **Yang, Z.**, L. Jacob, and L. Lewis, "Closed-loop auditory stimulation of sleep slow waves drives cerebrospinal fluid flow in humans," *Science Translational Medicine*, 2026.
- [5] **Yang, Z.**, S. Williams, E. Beldzik, S. Anakwe, E. Schimmelpfennig, and L. Lewis, "Attentional failures after sleep deprivation represent moments of cerebrospinal fluid flow," *Nature Neuroscience*, 2025.
- [6] J. Levitt, **Yang, Z.**, S. Williams, S. Espinosa, A. Garcia-Casal, and L. Lewis, "EEG-LLAMAS: a low-latency neurofeedback platform for artifact reduction in EEG-fMRI," *NeuroImage*, 2023.
- [7] **Yang, Z.** and L. Lewis, "Imaging the temporal dynamics of brain states with highly sampled fmri," *Curr Opin Behav Sci*, 2021.

ACHIEVEMENTS

Organization for Human Brain Mapping Merit Award	<i>2024</i>
Kavli Summer Institute in Cognitive Neuroscience Fellow, UCSB	<i>Summer 2022</i>

SKILLS

Human Neuroscience	EEG, fMRI, fNIRS, Eyetracking, physiological timeseries analysis
Artificial Intelligence	Deep Learning, Large Vision Models, classical machine learning
Programming Languages	Python (Pytorch, Tensorflow, Sklearn, Pandas, Numpy), MATLAB (EEGLAB), R, git, Bash, C/C++
Large-scale Computing Languages	SLURM, multi-gpu computing Mandarin Chinese (Native), English (Proficient), German (Basic), Japanese (Basic)
Others	Adobe Acrobat Pro, Qualtrics, Microsoft Word, Excel, Photoshop, Film Editing, Cinematography

SELECTED TALKS

Invited Symposium Talk at FENS Forum 2026	2026/07
Invited Talk at Gordon Sleep Seminar	2026/03
Invited Talk at ISTAART Neuromodulatory Subcortical Systems PIA “Tiny Dots” (Virtual)	2026/01
Invited Talk at Brain Clearance Meeting (Virtual)	2025/12
Invited Talk at Stanford Precision NeuroIntelligence Lab	2025/10
Invited Talk at Harvard	2025/06
NeuroLunch Talk at MIT	2025/03

PRESENTATIONS

- [8] N. Otic, K.-C. Wu, M. Renna, A. Rakymzhan, J. Kim, S. Bailes, **Yang, Z.**, Z. Starkweather, O. Kierul, H. Yun, C. Zambello, D. Salat, M. Franceschini, L. Lewis, and M. Robinson, “Overnight monitoring of electrophysiological and hemodynamic signals associated with glymphatic function using a combined nirs and eeg approach,” in *Clinical and Translational Neurophotonics*, 2027.
- [9] E. Beldzik, N. Cicero, D. Gomez, **Yang, Z.**, J. Iglesias, E. B, and L. Lewis, “Subcortical neuromodulatory circuits for joint regulation of sleepiness and vascular physiology,” in *ESRS Sleep Europe*, 2026.
- [10] E. Beldzik, D. Gomez, N. Cicero, **Yang, Z.**, J. Iglesias, and L. Lewis, “Subcortical dynamics during failures in maintaining alertness after sleep restriction in the human brain,” in *Sleep*, May 2025.
- [11] E. Beldzik, **Yang, Z.**, S. Williams, and L. D. Lewis, “Distinct spectral pattern of cognitive, drowsiness, and fatigue-related theta/alpha eeg activity during wakefulness,” in *Sleep*, 2024.
- [12] **Yang, Z.**, S. Williams, N. Tacugue, Z. Valdiviezo, J. Hua, T. Ly, M. Aon, I. Vinal, E. Schimmelpfennig, N. Leonard, R. Huang, D. Zimmerman, J. Yee, and L. Lewis, “SWADEE: A GUI-based Tool for Slow Wave Activity Detection via EEG and Eyetracking,” in *Society for Neuroscience*, San Diego, CA, 2022.
- [13] S. Williams, **Yang, Z.**, S. Anakwe, J. Licata, E. Schimmelpfennig, M. Bosli, N. Leonard, I. Vinal, M. Aon, Z. Valdiviezo, N. Tacugue, and L. Lewis, “Changes in osmolyte concentration and excitatory-inhibitory balance after 24 hours of total sleep deprivation,” in *ISMRM MRS*, 2024.
- [14] S. Williams, **Yang, Z.**, S. Anakwe, Z. Valdiviezo, N. Tacugue, I. Vinal, E. Schimmelpfennig, M. Aon, M. Bosli, J. Licata, N. Leonard, M. Ruiz, H. Fitzgerald, M. Otto, and L. Lewis, “Fast fmri imaging of amygdala bold hemodynamics in major depressive disorder after 26 hours of total sleep deprivation,” in *Society of Biological Psychiatry*, 2024.
- [15] E. Beldzik, **Yang, Z.**, S. Williams, and L. Lewis, “Elucidating the theta paradox: Distinct spectral characteristics of cognitive- and drowsiness-related increases in midfrontal theta eeg activity,” in *Society for Neuroscience*, 2023.
- [16] A. van der Kouwe, H. Jeong, **Yang, Z.**, D. Straney, R. Frost, L. Lewis, and G. Bonmassar, “The MotoNet: An MRI-Compatible EEG Net with Embedded Motion Sensors,” in *International Society for Magnetic Resonance in Medicine (ISMRM)*, 2022.
- [17] **Yang, Z.**, J. Pineda, I.-W. Shu, J. Onton, A. Rivas, N. Zhen, L. Ring, M. Bordyug, and F. Singh, “Update on a longitudinal pilot study to assess the effects of gamma neurofeedback on cognitive function in schizophrenia patients,” in *Society for Neuroscience Meeting*, San Diego, CA, 2018.
- [18] E. Herrera, F. Singh, **Yang, Z.**, L. Ring, A. Amello, and J. Pineda, “Role of gamma neurofeedback in working memory of persons diagnosed with schizophrenia,” in *Society for Neuroscience Meeting*, Washington, DC, 2017.
- [19] N. Dudeck, F. Singh, **Yang, Z.**, R. Cheng, R. Gosla, and J. Pineda, “Gamma neurofeedback synchrony training on working memory in schizophrenia,” in *Society for Neuroscience Meeting*, 2016.
- [20] F. Singh, A. Smith, **Yang, Z.**, and J. Pineda, “Neurofeedback on working memory in schizophrenia patients,” in *Society for Neuroscience Meeting*, San Diego, CA, 2016.

RESEARCH EXPERIENCE

Lewis Lab for Imaging Brain Dynamics

Supervisor: Dr. Laura D. Lewis

Sep 2019 - Current
MIT

- Conducted 100+ simultaneous fast fMRI-EEG-Eyetracking scans with human participants and developed analysis pipelines to examine the multimodal dataset.
- Led non-invasive cerebrospinal fluid (CSF) after sleep deprivation project with a team of 10: investigated the relationship between brain's neural signals and brain's fluid signal.
- Developed a MATLAB software (SWEET) for detecting local sleep activities in human with EEG and Eyetracking and presented the result at Society for Neuroscience.
- Implemented an AI-based generative model with PyTorch for predicting neural activity and behavioral performance from physiological signals (e.g., heart rate, pupil diameter, skin conductance).

Functional Neuroscience Lab

Supervisors: Dr. Jaime A Pineda and Prof. Fiza Singh

Jan 2015 - June 2019
UC San Diego

- Led a team of 5 undergraduate researchers to deliver EEG-based neurofeedback to treat schizoaffective disorders.
- Performed EEG data processing such as filtering, artifact rejection, source localization, time-frequency analysis, and independent component analysis.
- Utilized both the Cognionics Quick-20 dry-EEG electrode system and the wet electrodes to administer EEG recording and Neurofeedback training on schizophrenic and normal populations.
- Maintained and implemented experiments using stimulus presentation software such as Neurobehavioral System Presentation.
- Performed statistical analysis of neuropsychological test data using Excel and SPSS. Experience with Thought Technology ProComp Infiniti EEG Suite, Version 6.0 and EEG data analysis using EEGLAB and NeuroGuide.

de Sa Lab

Supervisors: Dr. Virginia de Sa

Jan 2018 - June 2019
UC San Diego

- Contributed to the development of BCI P300 speller experiments with Python-based game design package SNAP.
- Set up Brain Products EEG system and helped conducting BCI motor-imagery experiments.

Vision and Memory Lab

Supervisors: Dr. Timothy Brady and Dr. Mark Schurgin

Jan 2018 - June 2019
UC San Diego

- Investigated the capacity of visual long-term memory leveraging behavioral psychological methods.
- Led behavioral experiments involving recruitment of and guidance to participants to complete a series of computer-based tasks while recording their outputs for further comparison/analysis.
- Performed advanced data analysis in MATLAB and R to analyze test results, generate receiver operating characteristic (ROC) curve and calculated area under the curve (AUC).

AD HOC REVIEWER

Current Opinion in Behavioral Sciences, Communication Biology, Nature Progress Brain Health, BioMedInformatics, Journal of Interdisciplinary Research Applied to Medicine, Academia Medicine and Health